What does the Knowledge Graph have to do with the INTFOROUT project

Let's compare with the Google Knowledge Graph (Google KG) !

GOOGLE KG	INTFOROUT KG
Answer users who search for information on the Web (and advertisers who want to understand what readers are interested in)	Answer scientists who want to study the impact of human outdoor activities on the French Alps ecosystems using data and advanced analytical methods
Google answers with URLs (and snippets) of Web pages and information	Answer with references to data, tools, and information
Produced by Google using : search logs, algorithms, human power, existing graphs and data	Produce specific pilots/prototypes that showcase how a KG can provide better answers to a question, that are produced maually and then partially automated , with the goal of building a collaborative and open KG

We conducted a small experiment
to scope a INTFOROUT KG pilot
that would « better » answers « real » questions

Message sent to project INTFOROUT participants :

Questions for an IntForOut AI

We would like you to write down in this document the questions you imagine you might ask an IntForOut AI in the near future, an AI that would be familiar with the topics studied in the project, the data available in the studied areas and the associated tools. The aim is not to list all the questions the knowledge graph will need to answer, but to collectively focus on the prototypes we want to produce during the project.

Here are the questions proposed by the participants:

Question 1: What information on mountain practices can be found in existing databases?

Question 2: What information is available on the movement patterns of hikers in the two study areas?

Question 7: What is the current state of knowledge regarding this animal's spatial behaviour during winter, and how does this knowledge relate to the topographical features shown on my map?

Question 8: "At what times and during which behaviours/activities are these animals most sensitive to human disturbance, and when are disturbances most harmful to the animals' wellbeing?"

Question 3: Are there any animals that are often (or never) found at a distance of less than <a certain distance> from the trails? Near mountain huts or other points of interest?

Question 4: Which animals avoid the trails? Is this linked to visitor numbers?

Question 5: I have a trail of an animal; where are its resting spots, what explains them, could they be sources of interaction with humans, and what are the seasonal differences in animal use of the areas adjacent to trails?

Question 6: What is the level of human use of the routes depending on the season and time of day? Where are the areas where people stop and congregate? Can we identify different types of use depending on the route (heading for a mountain hut, off-trail, starting or finishing via ski lifts, etc.)
Translated with DeepL.com (free version)

The expected types of answers are not quite the same:

Requests for pointers to data or tools

Question 1: What information on mountain practices can be found in existing databases?

Question 2: What information is available on the movement patterns of hikers in the two study areas?

Requests for an article

Question 7 : What is state of the art knowledge about this animal' spatial behavior during the winter and how this knowledge relates to topographic entities displayed on my map?

Requests for a map or specific information

Question 8: "At what times and during which behaviors/activities are these animals most sensitive to human disturbance and when are disturbances most harmful to the wellbeing of the animals?"

Question 3: Are there any animals that are often (or never) found at a distance of less than <a certain distance> from the trails? Near mountain huts or other points of interest?

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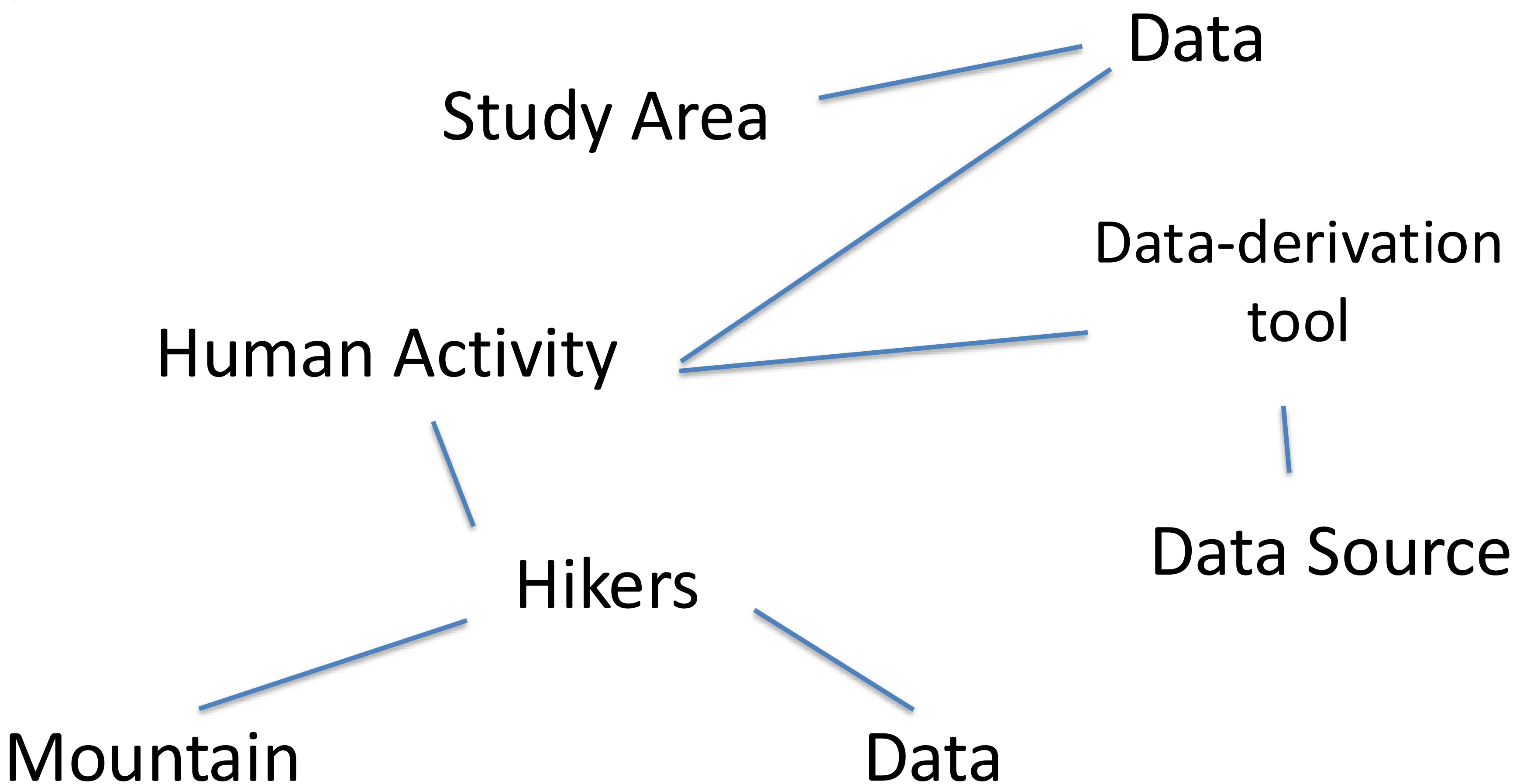
What should an INTFOROUT Knowledge Graph look like to answer requests* for data and tools ?



* Question 1: What information on mountain practices can be found in existing databases?

Question 2: What information is available on the movement patterns of hikers in the two study areas?

If a KG knows the existing data and tools, we can imagine that the prototype could also answer other real questions raised during the project meetings, such as:

What is the difference between these two data sources?

Is there a research dataset similar to mine that already has metadata?

What is the relevant distance between these animal GPS tracks and the topographic objects?

Other questions* involve the ability to take time into account more effectively

Style

Time

Data

Animals

Trail

Data

Human Activities

* Question 8: “At what times and during which behaviors/activities are these animals most sensitive to human disturbance and when are disturbances most harmful to the wellbeing of the animals?”

Question 6: What is the level of human use of the routes depending on the season and time of day?

An INTFOROUT KG that can answer with information and maps must be able to launch data queries, processing tasks, and visualizations.

Data Request Workflow
and Map Creation

Distance

Style

Time

Données

Animal

Trail

Données

Human Activities

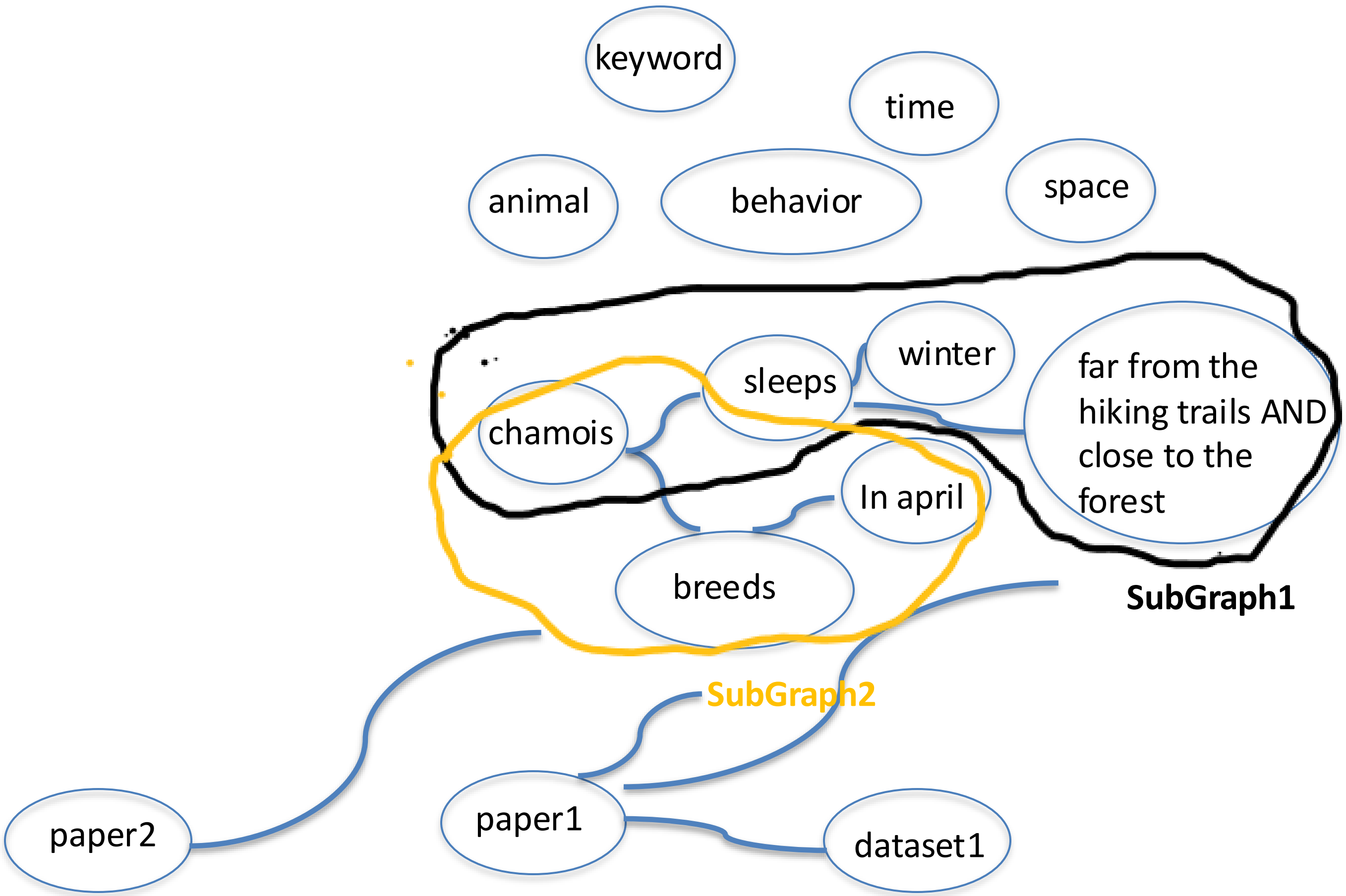
If a prototype is able to generate maps, we can imagine that it could also answer other questions, such as:

What colors should be used to represent these different themes on the same map?

Should I include reading helpers for these data and for this audience?

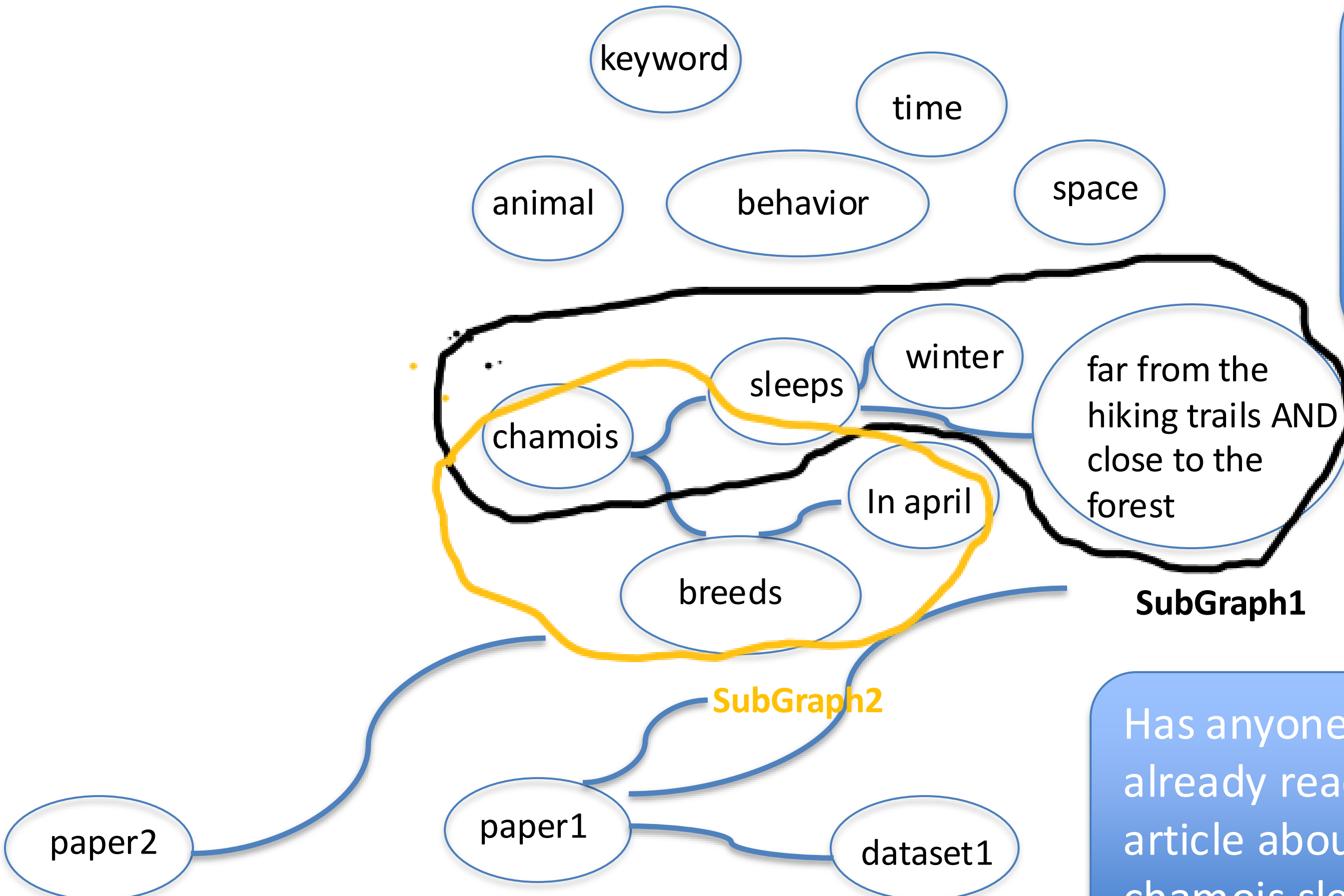
Which result should be shared for which recreational activity, at what time, and through which media?

An INTFOROUT Knowledge Graph could answer questions about the knowledge related to the phenomena being studied



Question 7 : What is state of the art knowledge about this animal' spatial behavior during the winter and how this knowledge relates to topographic entities displayed on my map?

Such an INTFOROUT KG could also answer other questions, such as:



Is this term used as a keyword in different articles, and what are the key references that define it?

Has anyone in the project already read and shared an article about the places where chamois sleep?

Scope of the 1st pilot : answer the question « What data exist about human activities in our study areas? »